

EE 175AB Senior Design · ENGR 181W Technical Communication

Battery Management System

A 10-Cell Lithium-Ion Pack with Programmable
Electronic Load and Real-Time Desktop Dashboard

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Commercial 10-cell BMS units cost \$1,200+ and lock telemetry behind proprietary software — none feed a custom Python application.

The gap

Safety: a 10s pack at 42 V can source 100+ A into a short — no MCU response is fast enough.

Cost: Orion BMS 2 starts at \$1,200 and requires Orion's proprietary configurator.

Integration: no commercial unit exposes per-cell voltage over a standard USB serial port.

Five design objectives

Cell voltage error $\leq \pm 20 \text{ mV}$

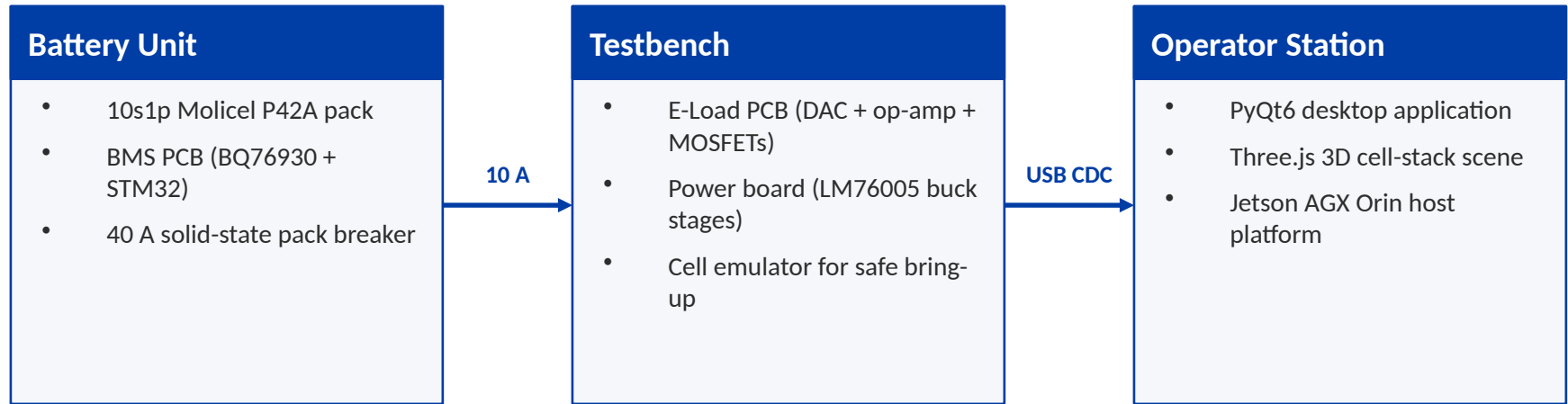
Breaker response $\leq 5 \text{ ms}$

E-Load regulation within 2 % of set-point

Dashboard refresh $\geq 5 \text{ Hz}$ live

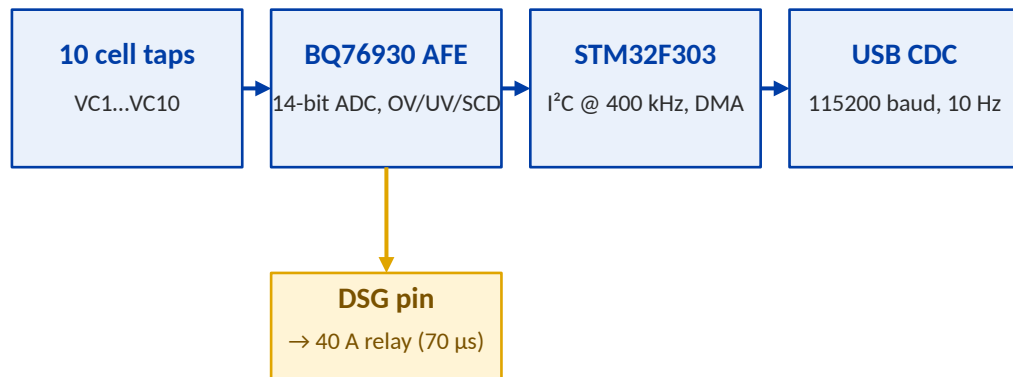
Total budget $\leq \$3,000$ across 4 PCBs

The testbench splits into three domains linked by USB CDC and a QWebChannel software bridge.



Both STM32 boards expose USB CDC virtual COM ports at 115200 baud · QWebChannel bridges Python → JavaScript at 10 Hz

The BQ76930 AFE opens the pack breaker in 70 μ s under short circuit — hardware-enforced, independent of firmware state.



What this buys you

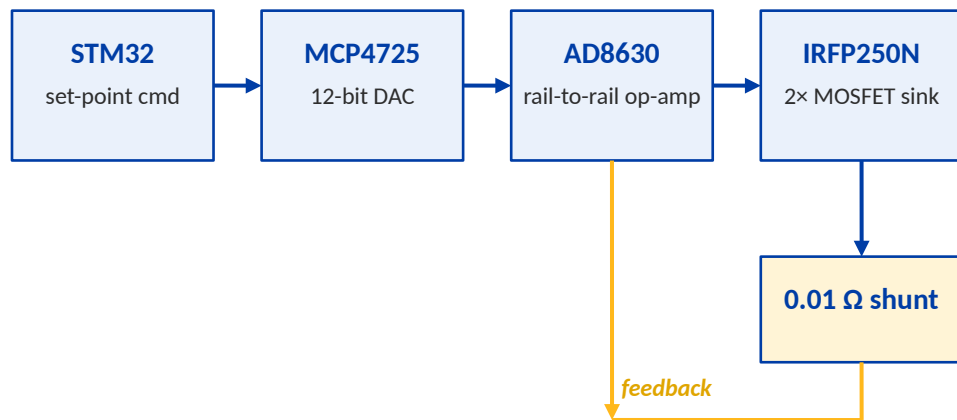
Autonomous protection fires even if firmware is mid-reset or stuck.

1.2 mV ADC resolution from a calibrated 14-bit per-cell ADC.

100-byte telemetry frame with CRC-16 over USB CDC every 100 ms.

Source: BQ76930 datasheet (Rev. H), Texas Instruments

A 12-bit DAC and AD8630 op-amp close the feedback loop that holds 10 A discharge within 2 % across the 30–42 V pack range.



Measured performance

1.5 % regulation across 36–42 V under step and ramp profiles.

200 W dissipation shared by two MOSFETs on a 0.12 K/W heatsink.

70 °C cutoff via NTC thermistor — 25 °C of thermal margin.

Source: E-Load bring-up data, EE 175AB test logs, May 2026

Three software layers stream 10 Hz telemetry from STM32 firmware to a Three.js scene with no network stack.

Layer 1 — STM32 Firmware (C / HAL)

DMA-backed I²C polling · 100-byte CRC-16 framed telemetry · ALERT interrupt for sub-cycle faults



Layer 2 — Python Backend (PyQt6)

QThread per CDC port · auto-reconnect on USB drop · QWebChannel exposes telemetry object to JS



Layer 3 — Three.js Frontend

10 cylindrical meshes (2×5) · green→yellow→red emissive mapping · GSAP fault state transitions

Why this stack

USB CDC over CAN removes external transceivers and adapters.

Qt WebEngine over Electron eliminates CORS and a 200 MB Chromium binary.

STM32 over Pi Zero delivers 100 ms USB frames with no scheduler jitter.

All five design objectives were met — three were exceeded by 2× or better against the original spec.

Objective	Target	Measured	Margin
Cell voltage error	$\pm 20 \text{ mV}$	$\pm 10 \text{ mV}$	2× better
Breaker response (SCD)	$\leq 5 \text{ ms}$	70 μs	70× better
E-Load regulation	$\leq 2 \%$	1.5 %	within spec
Dashboard refresh	$\geq 5 \text{ Hz}$	10 Hz steady	2× better
Total system cost	$\leq \$3,000$	\$966	32 % of budget

Headline: every quantitative spec passed — and the system shipped at 32 % of the allocated budget.

Conclusions

- 1. Hardware-enforced protection works:** 70 μ s short-circuit response with zero firmware involvement.
- 2. The integrated testbench beats commercial alternatives on observability** — per-cell voltages stream natively to a custom Python application.
- 3. All five objectives met under one-third of budget.** Active balancing and SOC estimation are the next milestones.

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References

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